

Lesson 32: The Normal Distribution — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. What shape is a normal distribution?
2. Where is a normal curve centred?
3. $P(X < \mu)$ for a normal?
4. % within 1σ of the mean?
5. % within 2σ of the mean?
6. % within 3σ of the mean?

Section 2 — Core practice

7. $N(50, 5^2)$: between which values do 68% lie?
8. $N(100, 10^2)$: 95% lie between?
9. $N(20, 2^2)$: 68% lie between?
10. Does $N(\mu, \sigma^2)$ list σ or σ^2 ?
11. $N(60, 4^2)$: 95% lie between?

12. $P(X > \mu)$ for a normal?

Section 3 — Challenge & exam-style

13. Heights are $N(170, 8^2)$ cm. Between what heights do 95% lie?

14. Apples are $N(120, 10^2)$ g. Estimate $P(X < 130)$ using the 68% rule.

15. $N(500, 50^2)$: roughly what proportion is below 450?

16. A machine fills bottles $N(330, 5^2)$ ml. About what % are between 320 and 340 ml?